WORKED SOLUTION

STEP I 1994, Question 5

Pure • Loci on a Parabola

STEP I 1994, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/94-s1-q5

The key idea

Turn the geometric condition into algebra first. A right angle at the origin means the position vectors $\overrightarrow{OP}$ and $\overrightarrow{OQ}$ are perpendicular, and for points on $y = x^2$ this collapses to a single neat relation between p and q. Both points R and T then have coordinates that are symmetric in p and q, so we write everything in terms of $p+q$ and pq and eliminate.

Solution

The condition for a right angle

Neither P nor Q can be the origin, or the angle would not be defined, so $p \neq 0$ and $q \neq 0$. The angle POQ is a right angle exactly when the scalar product of $\overrightarrow{OP} = (p, p^2)$ and $\overrightarrow{OQ} = (q, q^2)$ is zero:

$$pq + p^2q^2 = 0 \quad \Longleftrightarrow \quad pq(1+pq) = 0 \quad \Longleftrightarrow \quad pq = -1,$$

dividing by $pq \neq 0$. (Equivalently, the gradients p and q of OP and OQ multiply to -1.)

So the pairs we want are exactly $q = -1/p$ with $p \neq 0$. As p runs over the non-zero reals, the sum

$$s = p + q = p - \frac{1}{p}$$

takes *every* real value: given any s, the quadratic $p^2 - sp - 1 = 0$ has discriminant $s^2 + 4 > 0$, so it has real roots, and their product is -1, so neither is zero. We will need this to say that the loci below are complete curves rather than parts of them.

Part (i)

The midpoint of PQ is

$$R = \left(\frac{p+q}{2}, \frac{p^2+q^2}{2} \right).$$

Write $R = (x, y)$, so $p + q = 2x$. Using $pq = -1$,

$$y = \frac{(p+q)^2 - 2pq}{2} = \frac{4x^2 + 2}{2} = 2x^2 + 1.$$

Since $p + q$ takes every real value, so does x, and the locus of R is the whole parabola

$$y = 2x^2 + 1,$$

with vertex $(0, 1)$ and axis the y-axis. It is narrower than $y = x^2$ and lies entirely above it, since $2x^2 + 1 - x^2 = x^2 + 1 > 0$; this fits with R being the midpoint of a chord of a convex curve.

Part (ii)

Differentiating $y = x^2$, the tangent at P has gradient $2p$, so its equation is

$$y - p^2 = 2p(x - p), \qquad \text{that is,} \qquad y = 2px - p^2,$$

and similarly the tangent at Q is $y = 2qx - q^2$. These are not parallel, since $p \neq q$ (their product is -1, so they cannot be equal). Subtracting the two equations,

$$2(p - q)x = p^2 - q^2 \quad \Longrightarrow \quad x = \frac{p+q}{2},$$

dividing by $p - q \neq 0$. Then

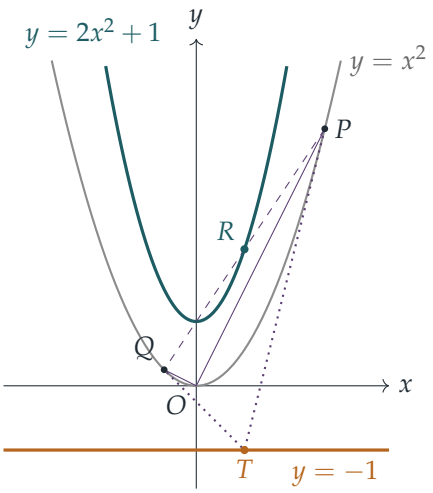
$$y = 2p \cdot \frac{p+q}{2} - p^2 = pq = -1.$$

Ask yourself The y-coordinate of T is constant. Is the locus a single point, a segment, or the whole line?

The x-coordinate of T is $\frac{1}{2}(p + q)$, which we showed takes every real value. So T can be anywhere on the line, and the locus of T is the whole straight line

$$y = -1.$$

Notice also that T always lies directly below R: both have x-coordinate $\frac{1}{2}(p + q)$.



The figure shows one position, $p = 2$, $q = -\frac{1}{2}$: here $R = (\frac{3}{4}, \frac{17}{8})$ on the teal parabola and $T = (\frac{3}{4}, -1)$ on the amber line.

Checks

Take $p = 1$, $q = -1$: then $P = (1, 1)$, $Q = (-1, 1)$, and OP, OQ are the lines $y = \pm x$, clearly perpendicular. The midpoint is $(0, 1)$, the vertex of $y = 2x^2 + 1$, and the tangents $y = 2x - 1$ and $y = -2x - 1$ meet at $(0, -1)$, on $y = -1$. With $p = 2$, $q = -\frac{1}{2}$ as in the figure, $2 \cdot (\frac{3}{4})^2 + 1 = \frac{17}{8}$, as it should be. A numerical check over 10^5 random values of p confirmed that R satisfies $y = 2x^2 + 1$ to within about 10^{-10} and that T always has $y = -1$.

Common slips

- Stopping at $pq(1 + pq) = 0$ without explaining why $pq = 0$ is ruled out: it would put P or Q at the origin.
- Giving the equations of the loci without checking that every x is attained, so that the sketch is of the whole curve rather than part of it.
- Assuming T lies on the directrix. The directrix of $y = x^2$ is $y = -\frac{1}{4}$; tangents meet there when PQ passes through the focus, which is a different condition from $\angle POQ = 90°$.

Beyond STEP

The chord PQ here always passes through the fixed point $(0, 1)$: its equation is $y = (p + q)x - pq = (p + q)x + 1$. Families of lines through a fixed point, and the matching "poles" where tangents meet (here on the line $y = -1$, the polar of $(0, 1)$), are the start

of projective geometry, where conics and their pole–polar relationships are studied systematically. For more problems in this spirit, try Siklos, *Advanced Problems in Mathematics*.

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